

*Zooming “In” and “out of” urban centers
in a rapidly developed metropolis:
A Isobenefit lens*

Juhyun Lee

Xi'an Jiatong-Liverpool University
Urban Planning and Design Department

Future mobility planning: AVs, Infrastructure, TODs...

Increasingly, mobility planning concerns “experiential value” beyond connection

- **Daily experiential values** (psycho-economic distance) vs. **time/distance to transit** (D'Acci, 2019)
- “City is NOT just hyper connected space....:
Legitimacy lies **heterogenous** citizens “**feeling of belonging**” culturally, spatially, socially within the **flow** (Bauder, 2016)

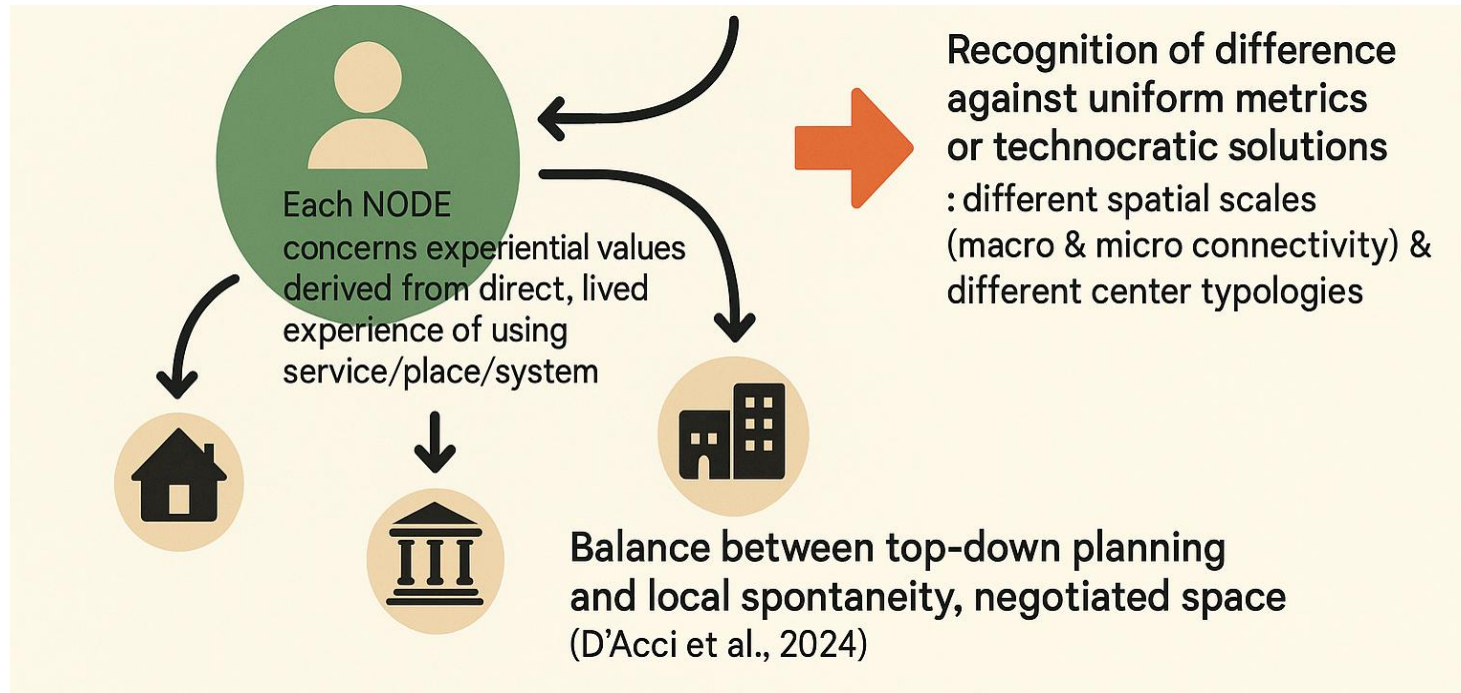
Two birds to catch: Contextualities & territorial balance matter (Hongkong, Seoul, Tokyo...)

- **Transit planning concerning scales** (e.g. HSR) vs. **Diverse hubs(nodes) across neighborhoods**
- Which **areas(nodes)** have/lacks experience **rich, service-oriented, daily life experience??**

Re-thinking Isobenefit Urbanism

Urban space and infrastructure to be organized in a way - Individuals *experience fair access* to urban benefits (amenities and nature) (D'acchi, 2019)

- **Territorial Balance:** Balanced distribution of urban benefits across city across locations
- **How:** strategic **macro planning** combined with **micro evolution** (D'Acchi et al., 2024) - "Third way"
- * **Direct, lived experience of** using service/place/system at **each node** as liquid surface rather than a solid"



=> **Contextuality**
over uniform metrics
or technocratic
optimization only

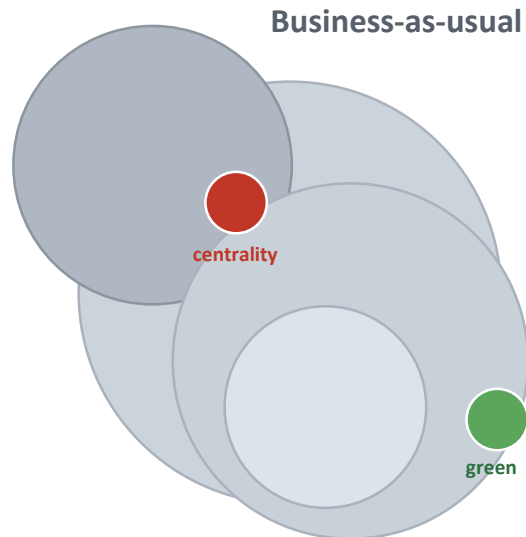
Scalable proximity: From Local Closeness to Multi-Centre Balance

Links two complementary—but non-substitutable—conditions: regional connectivity & local proximity

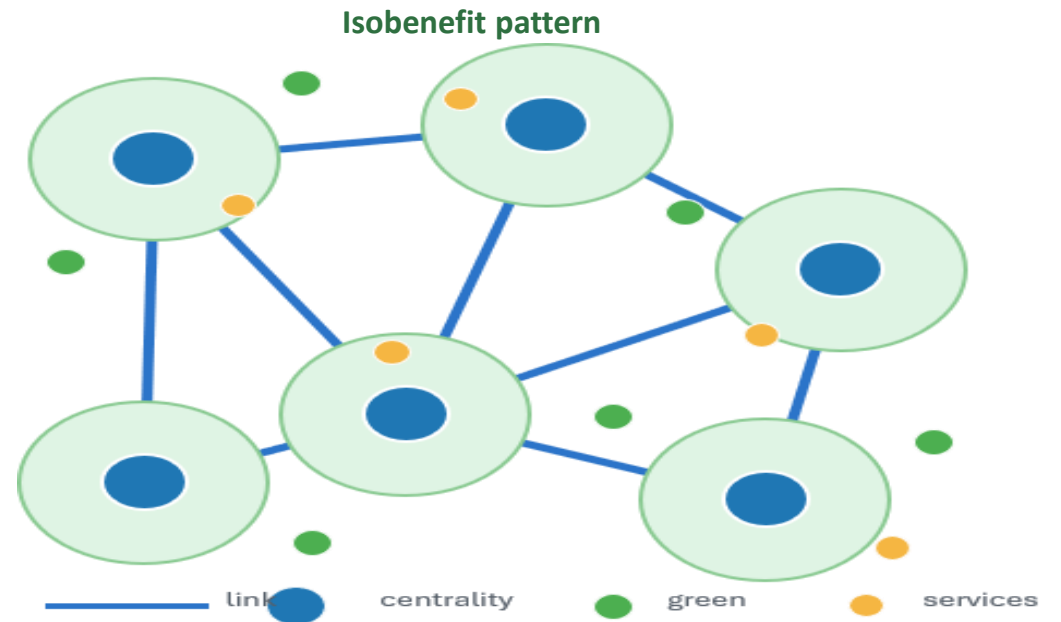
Extending the principle of everyday closeness **from individual neighbourhoods to a network of interconnected urban centres across the wider city-region** (Arias et al., 2025; Bruno et al., 2024)

- **Micro:** everyday services and nature within walking or cycling distance
- **Macro:** connections between centralities and wider metropolitan opportunities

Territorial balance across scales: Urban growth is organised around **multiple service-rich centralities**, each providing local access to everyday amenities and green space while remaining connected to other centres through public transport.



As the city expands, edge residents become farther from services and nature.



“Scalar tensions” at urban centers in compact polycentric cities

Scalar tension of (designated) in rapidly developed cities

City-wide transit connectivity vs. Everyday Accessibility

MACRO SCALE

City-wide connectivity

How strongly is the node positioned within the metropolitan public transport network?

Typical evidence:

centrality • ridership • travel-time role

MICRO SCALE

Everyday accessibility

Can people use services, amenities and public space comfortably in daily routines?

Typical evidence:

walkable services • capacity • comfort • barriers

A center (hub) may succeed at one scale and fail at another (e.g.)

*e.g. Central business districts (CBDs)

MSI: Zooming & out of urban centers/transit hubs across Metropolis

A center-sensitive diagnostics within polycentric (multi-centered) planning systems

- Jointly assessing two “non-substitutable” dimensions: city-wide public transport connectivity and Local service accessibility
- Interpret scalar coherence: Centers, as relational unit, read through alignment, mismatch and center-specific intervention

* Moving beyond aggregate urban-form measures, general accessibility metrics, and isolated scalar analyses.

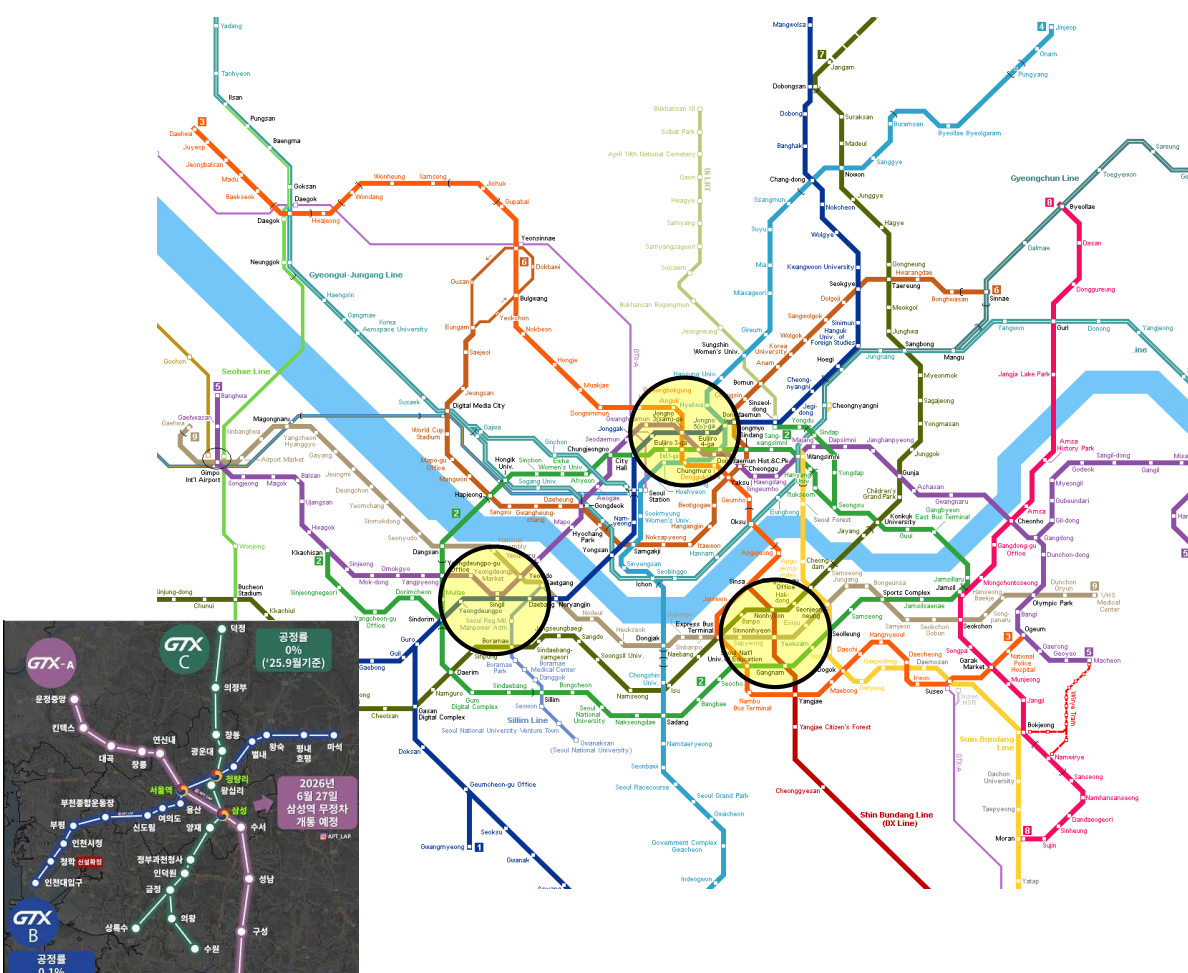
Dimension	Normative proposition in <i>isobenefit</i> urbanism	MSI operationalization	Evaluative logics	Selected indicator(s)
 Macro-scale: City-wide connectivity	Interconnected centralities	Relative positional importance of each center within the metropolitan public transport network (“zooming out”)	• High values: strong relative integration into city-wide opportunity • Low values: relative isolation or marginality from city-wide benefits	Time-weighted Betweenness centrality (BCT); Passenger-weighted PageRank centrality (PRP)
 Micro-scale: Local service accessibility	Equitable daily life access	Walkable or cyclable access to essential services and public amenities around each center (“zooming in”)	• High values: high and balanced accessibility within walking distance, with diverse service types available close to the node • Low values: service scarcity, distance burden, or mono-functionality	Gravity-based accessibility index using distance thresholds (400/800/1600m) and facility capacity (m²) across service categories
 Joint condition (<i>Isobenefit</i>)	Simultaneous access to macro and micro urban benefits	Parallel assessment and comparative interpretation of macro and micro performance for each center	• Convergence: simultaneous achievement of high scores in both dimensions indicates <i>Isobenefit</i> status • Divergence: significant gap between macro and micro scores indicates scalar imbalance or disintegration	Gap analysis across both dimensions, within and across different centers

MSI application for compact multi-centered policy Seoul

Seoul planned as High Density Multi-Nucleated Mega city to facilitate “**balanced distribution of urban functions**”

-Policy-designated Center (**Key nodes in metro networks**) to provide key socio-economic activities, cultural, public services:

-Public transport accounted for more than 60% of all trips



- Policy designated Center, strictly guided by formal planning criteria: centrality, transit infra and services..
 - Level 1 (CBD), Level 2 (Quarter center/regional center), Level 3(District center)
 - Each center: different centrality within the SMG, location, development potentials, coverage of residential areas, available transit infra and services

Varying characteristics of localities at different urban hierarchy levels in Seoul (SMG, 2006).

Centres classified according to the urban hierarchy	Relative amount of foot traffic	Relative amount of commercial land use	Intended role	Diagnostic question
Level 1	High	High	Metropolitan/global hub	Is high connectivity matched by local liveability?
Level 2	Middle	Middle	Regional balancing hub	Has it consolidated as a vibrant centre?
Level 3	Low	Low	Everyday local centre	Does local accessibility align with network role?



the land use within these centres is functionally mixed, with residential activities embedded even in commercial intensive areas such as the CBD—a pattern commonly observed in high-density Asian metropolises (e.g., Zheng et al., 2021; Lau et al., 2005)

Whether and how “planned urban centres (transit hubs)” simultaneously deliver metropolitan connectivity and local everyday accessibility, particularly at the level of individual policy-designated hub as units of intervention?

Diagnosis of macro-micro scalar tension within and across selected policy-designated centres

"how macro-micro configurations may differ across the selected cases?"

* One node purposively selected from each level center: 1 in CBD, 1 in Regional, 1 in District center

Each center (hub/node)'s configuration

Macro: Networked opportunities

"City-wide connectivity via transit"

Q: To what extent a node (key node) is connected to the whole transit network?

Measured using Network centrality metrics

(Between C + Degree C – time/ passenger weighted).

Micro: Local service in daily routine

"Local-service accessibility"

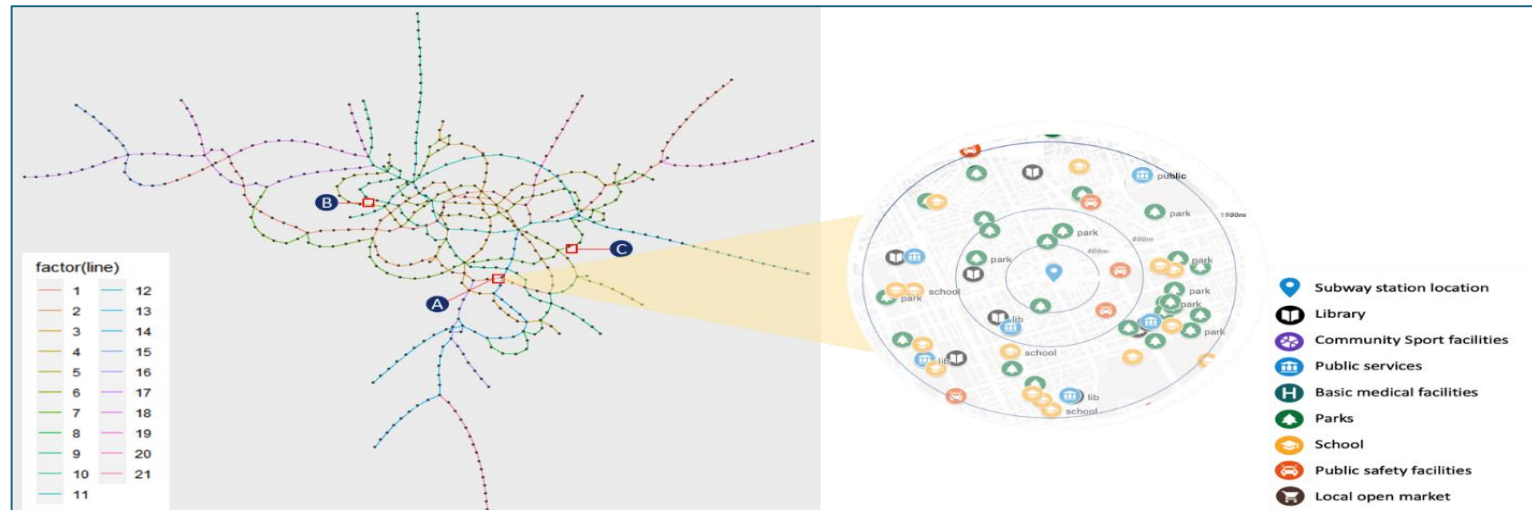
Q: To what extent urban services are usable/ accessible in daily routine?

Measured using Spatial facility data gravity model

(number/ capacities of service) (buffer: 400m/800m/1600m)

Objective
+
Subjective
measures

Validate experiential values (Urban quality of life & barriers to use service)



A simplified multi-scale investigation on centers

MSI evaluation of experiential values

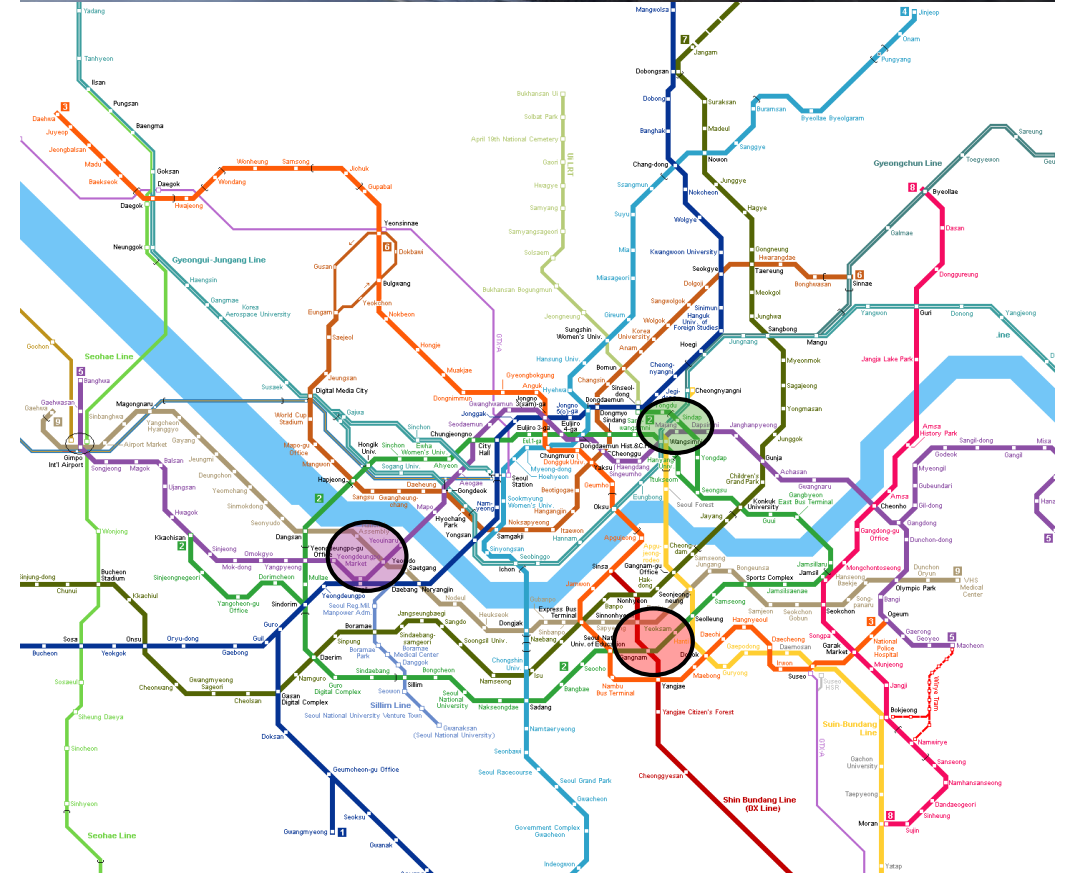
Node A in CBD

Node B in Regional/ Quarter Center

Node C in District Center

82,666 (A), 90,600 (B), and 88,311 (C) are broadly comparable in scale

- 8 focus groups with *elected local representatives* (of each center) “+ 15 years”:
Experiences/ perceptions on networked opportunities & local service accessibility **Change**
- Group discussions with **planners (SMGs) and local authorities** in charge of each node:
: land use patterns change
- Observations: **vicinity** of the selected nodes to observe the quality of BE and pedestrian path



City-wide connectivity: networked opportunities at selected nodes

CBD and District perform important bridging roles, while the Quarter Centre remains weakly consolidated within the metropolitan network.

Level 1(CBD): Rank highest in both measures - confirming bridge role/ highly functional

Level 2 Center: Lowest rank/ imbalance - need to enhance transit-land use mix

Level 3 Center: Balanced



CBD (A)

Metropolitan gateway

BCT top 7.31%

PRP top 1.92%

Highest overall network prominence; strong structural & actual-use centrality.



Quarter Centre (B)

Weak network consolidation

BCT top 34.81%

PRP top 17.88%

Lowest BCT and PRP; intended regional-hub role not fully realised.



District Centre (C)

Strong intermediary role

BCT top 8.08%

PRP top 14.04%

Strong structural brokerage; less intensive usage than CBD.

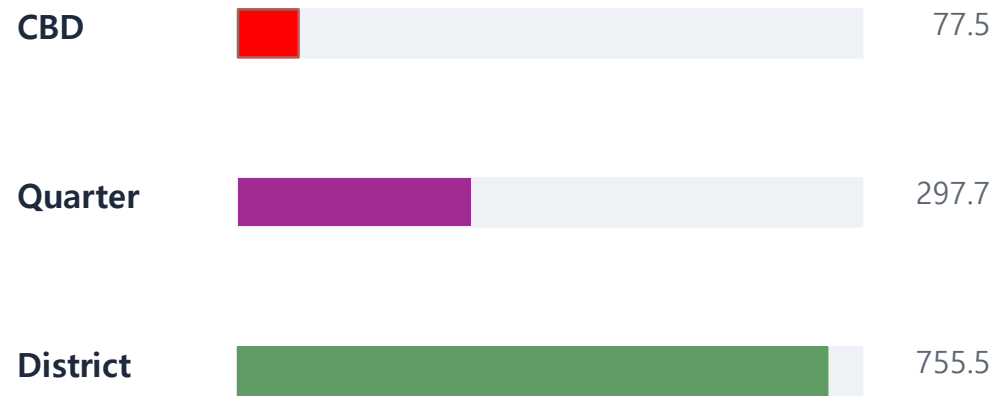
Local service in daily routine (local service accessibility) at selected nodes

CBD: lowest walkable accessibility; many essential services sit beyond practical walking range (with High satisfaction)

Quarter: strong immediate 400m provision, but only moderate overall accessibility.

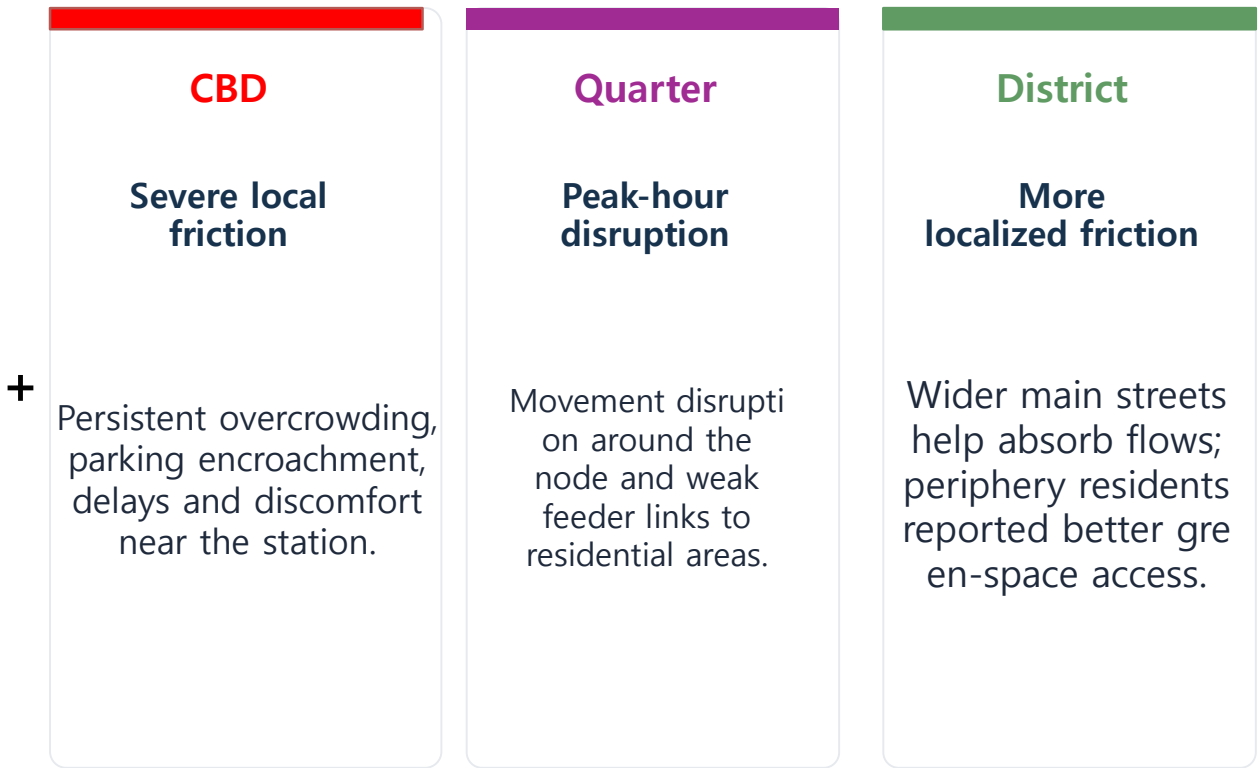
District: highest overall and walkable accessibility; most compact everyday-service structure.

Walkable service access (0–800m)



District 1st · Quarter 2nd · CBD 3rd

Differentiated level of local mobility friction



Experienced accessibility: Distance ≠ usable access

- Scalar tension between Macro vs. Micro scale
 - Macro: Clearly perceived benefits from **enhanced city-wide connectivity** vs. Micro: Low experiential value
"increasingly limited daily services within walkable zone x high traffic as barriers to daily routines "
- Experiential value are not guaranteed by solely "networked opportunities": **Level 2/3 better support daily life**

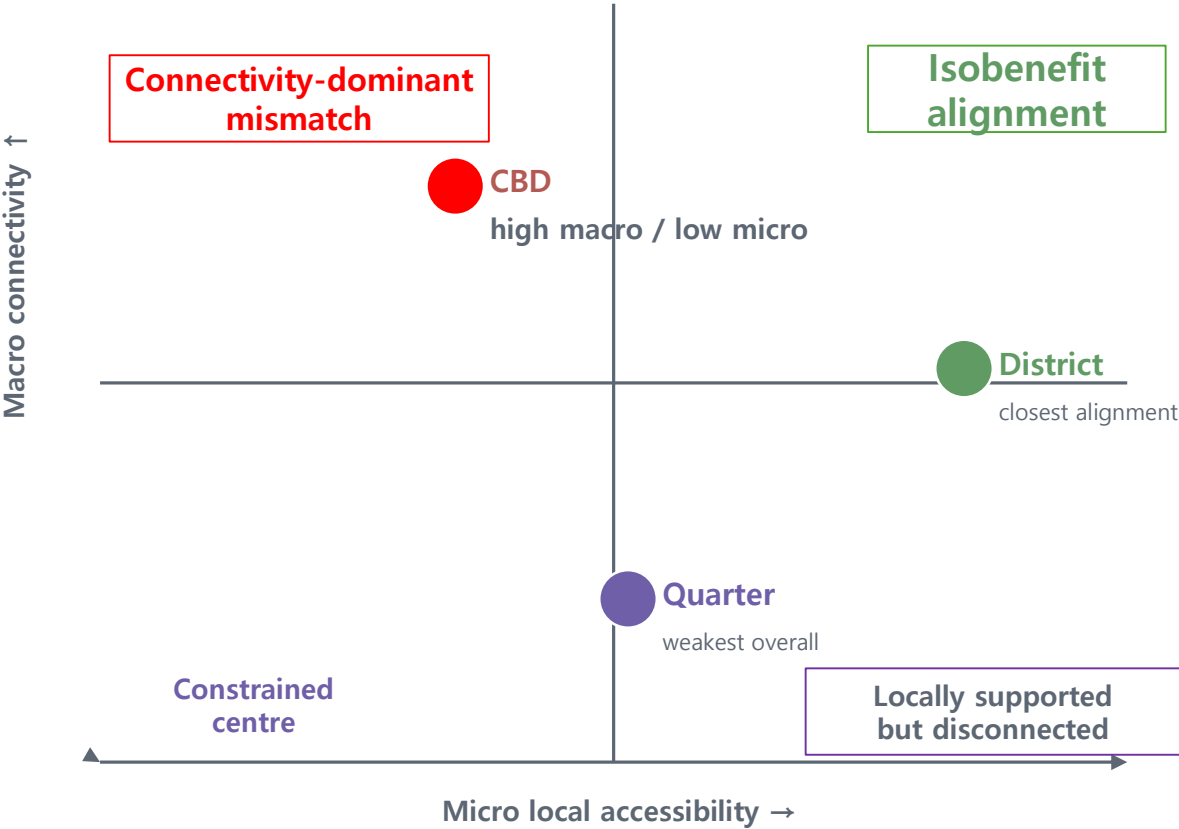
Extent of selected benefits arising from mega urban transport projects.

	Access to jobs	Access to local amenities	Quality of daily life	Distributional effects	Overall satisfaction
Level 1	High High (except for deprived areas)	Low High (except for deprived areas)	Low Moderate	Low High	Moderate High
Level 2					
Level 3	High	Moderate	High	Low	High

Three macro-micro configurations

The important finding is the type of mismatch, not a simple winner/ ranking.

- CBD (Level 1): Not winner, extreme mismatch** - Gateway function without corresponding everyday accessibility.
- District (Level 3): Highest livability integration** - closest approximation to the joint condition
- Quarter (Level 2): Most pronounced shortfall in the MSI evaluation** - Partial local function, weak metropolitan integration



District
Relatively coherent: strongest overall MSI profile and closest approximation to the joint condition.

Center	MSI	Networked Opportunities	Local Service in daily Life (walkable zone)	Experiential value
Level 1 (CBD)	Medium–	High	Very Low	Transit-rich BUT Pool experiential value
Level 2	Medium - Low	Medium	Medium	Local mobility inequity
Level 3	High	High	High	Transit-rich & Service – oriented, High experiential value

MSI changes the question from “which centre ranks highest?” to “what kind of mismatch exists here?”

Which quadrant describes your main station area?

Reflections on planning implications

- Strong city-wide connectivity did not consistently coincide with comfortable everyday mobility
- **Centre hierarchy does not predict lived performance:** Centre role, land use, transport position, and congestion conditions generate different scalar configurations.

CBD

**Connectivity-rich,
everyday-life poor**

Metropolitan gateway,
but local access can be
undermined
by congestion and weak service mix.

Mitigate local externalities

pedestrian-priority redesign • curb management • service protection

Quarter

**Partial local role,
weak hub consolidation**

Neither fully connected as
a city-wide hub nor
strong enough as a self-sustainin
g local centre.

Strengthen connectivity-building

feeder services • DRT • mobility hubs • last-mile networks

District

**Most coherent
macro-micro profile**

Stronger routine integration: lo
cal service accessibility and ne
twork role reinforce each other.

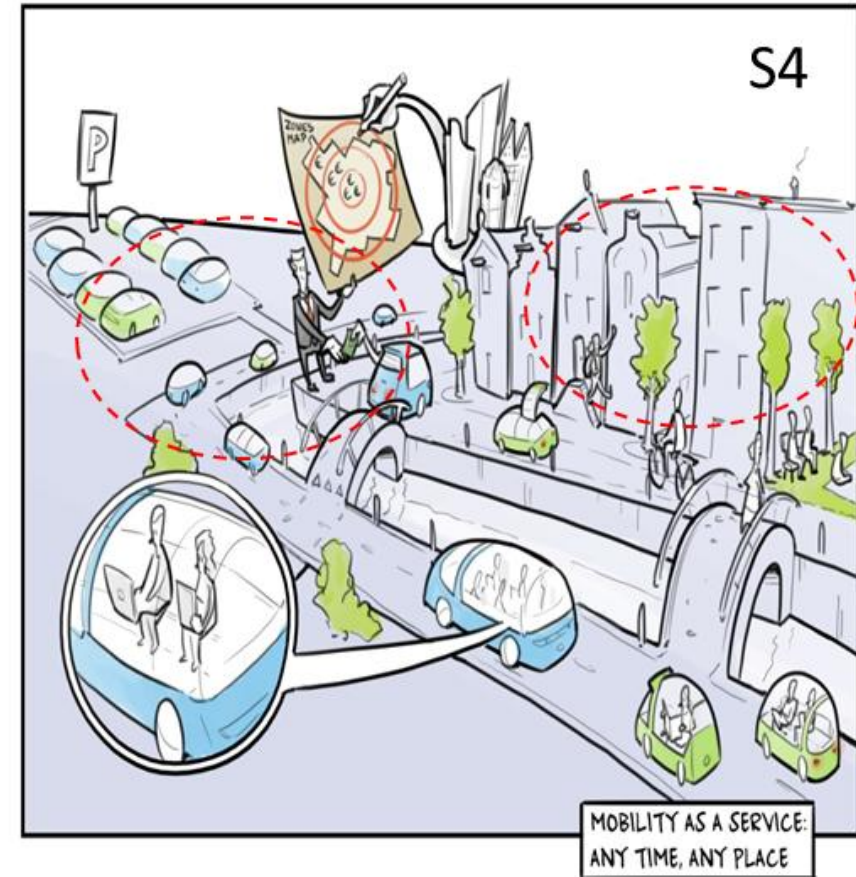
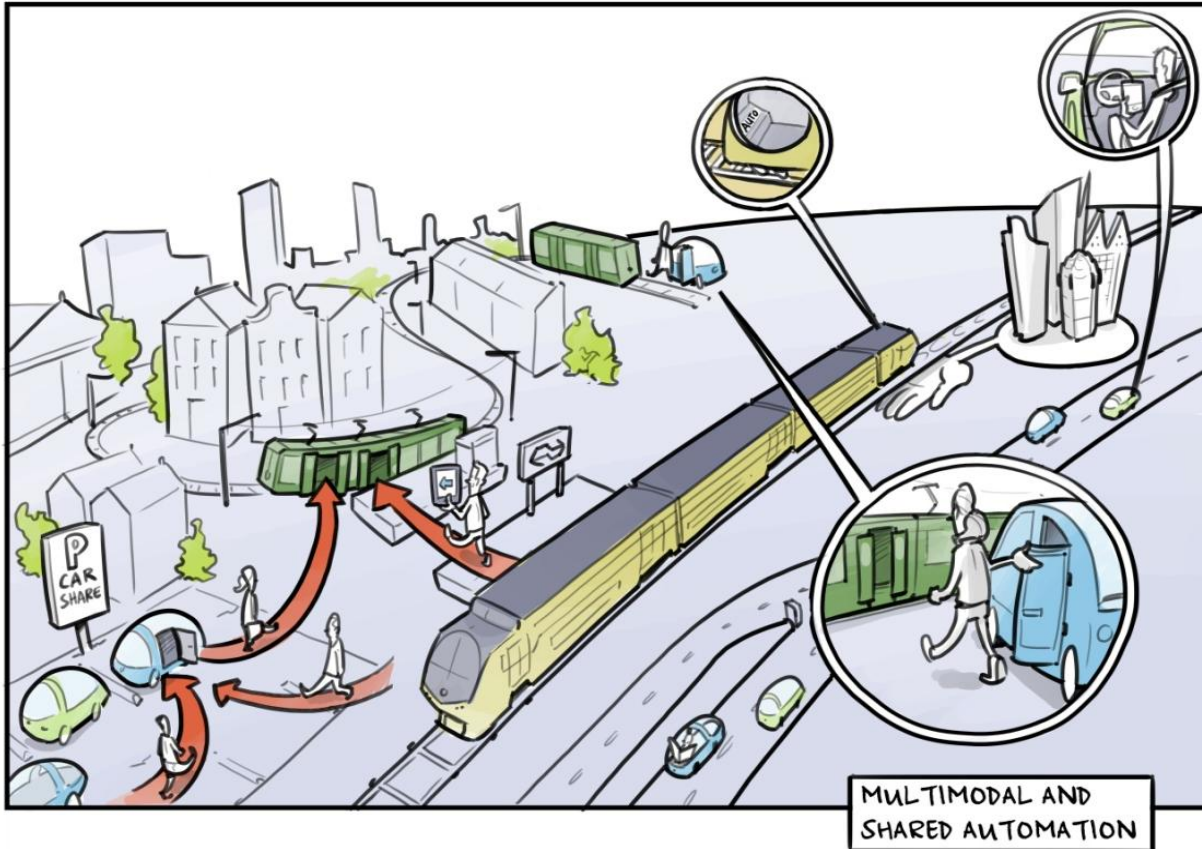
Preserve balance

traffic calming • pedestrian safety • service diversity protec
tion

Same framework, different adjustment logic.

Reflection on Socio-spatial implications of future mobility

The real challenge lies in ensuring that future mobility contribute to experiential urban quality and territorial balance across city



Extending the transit planning concept to include digital, spatial, and social practices

- Hierarchy within flow x networks of space
- Differentiate node typologies in transit planning for Mega cities/regions
 - : Different social and spatial functions
 - : Different strategies
 - : Transit-rich x poor experiential value
 - modular amenities; micro networks redesign; shared mobility hub
- Integrate TOD into shared urban experience
 - : invisible local nodes as everyday hub

Reference

Lee, J., Lee, J., Kim, MH. et al. (2026) Zooming in and out of compact urban centres in a rapidly developed metropolis: a multi-scale investigation through an isobenefit lens in Seoul. *Humanities and Social Science Communication*. <https://doi.org/10.1057/s41599-026-08270-0>

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Thank you

Juhyun.lee@xjtlu.edu.cn

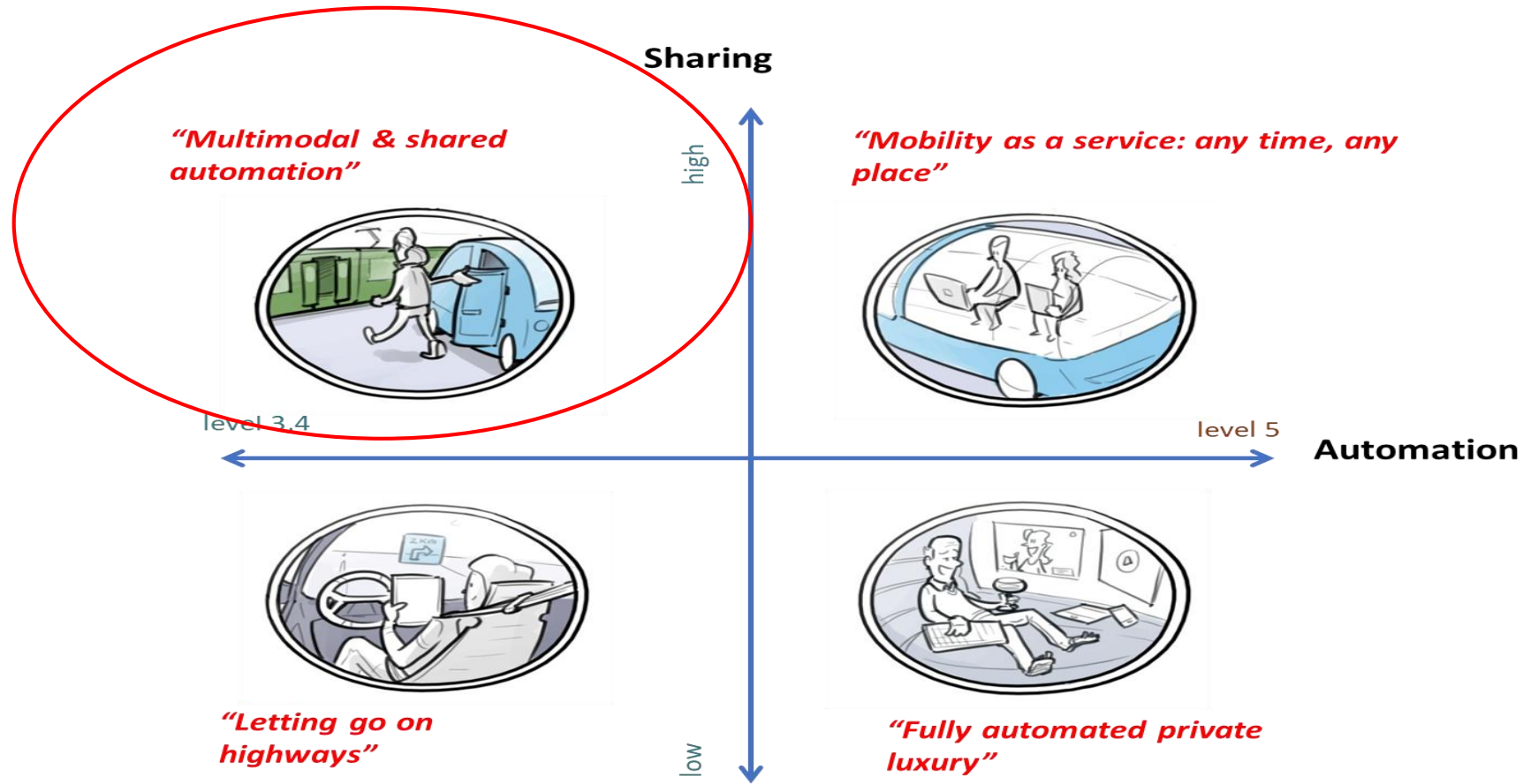
Contentexual sensitive X-TOD

- Contextualizing TOD Analysis and Planning
 - Locational/ functional context:
 - Transit technological context:
 - Policy and societal context

** Equitable tod study

Future mobility & territorial balance?

The real challenge lies in ensuring that future mobility contribute to experiential urban quality across city/ region.



(Dianin et al., 2021; Stead and Vaddadi; Tillema et al., 2015)

Re-thinking Isobenefit Urbanism

Urban space & infrastructure to be organized in a way: *Individuals* experience fair access to *livability opportunities*

- Recognition of **difference in terms of scales** (macro & micro connectivity) & **center typologies**
- Not optimization of time, but **balance of urban benefits** (Vs. Isochrones)

